

encryption and authentication cannot defend your network against clients that have already gained access to the network (compromised nodes or rogue access points), which often carry the private keys and other sensitive information.

Any structure or network is only as strong as its weakest link. Even after application of all security measures, we cannot safely say that the threats have been eliminated. Intrusion detection is an add-on security to the networks that help in forming the second line of defence against the potential attacks, and is an important addition to the networks that want to stay secure.

What's an Intrusion Detection System?

In simple words, an IDS or an Intrusion Detection System, is software that is used to detect unauthorised access attempts to a computer, mainly through network access. IDS does not, or rather, cannot prevent an attack, as many misguidedly believe. IDS only detects an attack being made in real-time. This means, when an IDS detects an intruder, some part of the network has already been cracked into and the attacker is already a part of the network.

Similarly, it is also not a firewall. It will not block traffic into your network. Though one can program the IDS to take some specific action if an intrusion is detected, such actions will be taken by some other application working together with the IDS.

A network IDS will continually monitor packets on a network and attempt to discover whether a system breach is in progress. Loss of information is not the only problem in case of wireless intrusion (or any network intrusion, for that matter). Some attackers may try to freeze your system, launch bomb attacks or try to cause DoS (Denial of Service) attacks on the system.

Some IDS systems can be tweaked to do some specific tasks upon finding an intrusion, such as finding the IP address and blocking it by invoking the firewall or finding the MAC address of the rogue machine and tracking its location. What must be noted here is that both the IP and MAC addresses can be spoofed easily.

So given that this is only a monitoring system, one needs to constantly monitor the network to see if an intrusion is happening. It is like having any other security system. You may fit a camera or a tracking device at your home to monitor unauthorised activity, but unless you actually see the activity in real time, you cannot prevent it. Watching recorded tapes is only the evidence that something bad has happened. You cannot go back in time and undo the damage. Your valuables are gone.

How wireless IDS works

As already discussed, a wireless network is more prone to attacks because it eliminates the need to connect a computer physically to an access point. This is the main reason why wireless IDS has to be different from traditional wired IDS. The main reason for this is explained below.

All wireless networks currently use the 802.11 IEEE standard of communication. So, when a client tries to communicate with an access point, both share the same frequency from a particular band in the range allowed by the 802.11 standard. They are also modulated by the same carrier

for proper communication. The 802.11 standard makes use of radio waves for modulation and communication.

Currently, wireless networks can operate in three main frequency bands: 900 MHz, 2.4 GHz, and 5.8 GHz. If you consider conventional FM radio, it is divided into a number of channels. We tune to specific channels like 98.3 and 94.1 MHz. Similarly, a wireless network uses a type of modulation known as FHSS (Frequency Hopping Spread Spectrum). The 2.4 GHz range is broken down into 79 different channels. It also uses the DSSS (Direct Sequence Spread Spectrum) type of modulation. Here, the 2.4 GHz range is broken into 11 channels. The available range is simply sub-divided into smaller parts to accommodate more channels or more simultaneous data transfer socket connections.

The reason I've made readers slog through the dreary math in the last paragraph is only to tell you that there is no single specific frequency in use at a given time. An IDS has to scan all the available channels in all the frequency bands available to sense the channels communicating at that time. So, even though the frequency range is standardised by the 802.11 standard, the detection is still very difficult. Therefore, the IDS must be able to scan all the channels, and that too, at a rapid rate so as to avoid loss of information regarding the intrusion.

We can configure an IDS to scan frequencies in only the band we use—802.11 a, b, g or n. To achieve this, the IDS uses a hopping system. (This is similar to frequency hopping in radio broadband communications and is beyond the scope of this article.) So, even though an IDS cannot collect all the data from all frequencies, it can only listen for the data from a channel when the frequency matches with that channel. During the remaining time, it cannot listen from that particular channel. After scanning all the remaining channels, it comes back to the initial one. (This is similar to time division multiplexing used in radio communications, again a subject that's beyond the scope of this article.)

Tracking rogue access points using Kismet

Irrespective of what kind of intrusion is taking place in a network, if the source of the problem can be found, then one can significantly eliminate the danger. A rogue AP (access point) is a Wi-Fi access point that is set up by an attacker. The purpose of a rogue AP is to sniff wireless network traffic in an effort to gain unauthorised access to the network environment under consideration. Once the attackers are in the network, they can disable services causing DoS attacks, collect sensitive information, or intercept secure data to be decrypted at their leisure.

There are many tools available to detect a rogue AP under Linux. A few of them are: Snort, Kismet and wIDS. Let's have an overview of rogue AP tracking with Kismet.

Kismet has all of the features that a normal packet filter has, but it also has many features dedicated to wireless networks. It has a built-in mechanism designed to detect any machine that's running NetStumbler, a program to gain access into a network. The software is also designed to decode WEP packets on the fly, as they are being captured. Crackers,